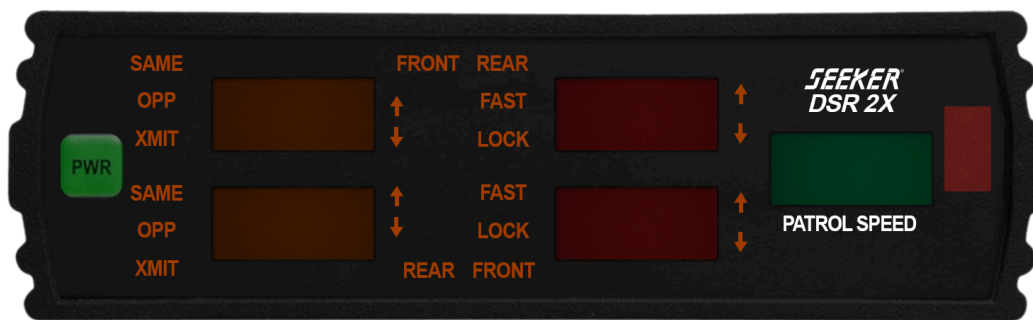


SEEKER DSR 2X

DUAL-ANTENNA TRAFFIC RADAR SYSTEM

Operator's Manual



This manual covers operation of the **seeker_dsr2x** radar system for FiveM: the display unit, the hand remote, the operator menus, the plate reader, and every setting available to the operator. Sections marked *server owner* describe values that are set in the resource configuration and cannot be changed from inside the game.

Resource: seeker_dsr2x • **Version:** 1.0.0 • **Author:** lwkjacob • **Requires:** ox_lib

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1. INTRODUCTION

The DSR 2X is a two-antenna traffic radar. One antenna faces forward and one faces to the rear, and **both measure at the same time**. The display is split horizontally to match: the top half belongs to the front antenna, the bottom half to the rear antenna, and the green PATROL SPEED window on the right is shared by both.

There is no antenna-select key on this unit. Anything you can set on one antenna you set separately on the other, using that antenna's own row of keys.

1.1 How the radar system works

Each antenna watches a beam that runs forward and backward along the patrol vehicle. Every vehicle inside that beam produces a return. The strongest return is the one shown in that antenna's **TARGET** window.

Return strength depends on how large a vehicle is, how far away it is, and how far off the centre of the beam it sits. Distance matters most by a wide margin, so a car close to you will normally read ahead of a lorry much further down the road — but at similar distances the larger vehicle wins.

If a vehicle in the same beam is travelling **faster** than the one in **TARGET**, its speed appears in that antenna's **FAST** window. This is the classic case of a car overtaking a lorry: the lorry gives the stronger return and holds **TARGET**, while the car's higher speed is pulled out and shown in **FAST**.

NOTE

FAST is a second reading taken out of the same beam, not a second beam. It follows whatever **TARGET** is doing, and it has no audio tone of its own.

1.2 Target zones

Each antenna is told which traffic to accept using its two zone keys, **OPP** and **SAME**. What they mean depends on the operating mode:

Zone key	Moving mode (MOV)	Stationary mode (STA)
OPP	Opposite lane — traffic coming the other way.	Targets closing on that antenna.
SAME	Same lane — traffic going your way.	Targets moving away from that antenna.

At least one zone must be selected or that antenna produces nothing. In stationary mode both may be selected at once, which makes the antenna bi-directional.

IMPORTANT

In stationary mode the zone keys are *mirrored* between the two antennas. A vehicle approaching your front bumper and a vehicle approaching your rear bumper are travelling in opposite directions, so "closing" is a different heading for each antenna. The unit handles this for you — select **OPP** on both rows and both rows will show traffic coming toward you.

2. CONTROLS AND COMMANDS

2.1 Default key binds

All key binds can be changed in **Settings** → **Key Bindings** → **FiveM**.

Key	Action
F5	Open and close the remote control.
ESC	Close the remote.
NUM 8 / NUM 2	Speed lock / release — front antenna / rear antenna.
NUM 9 / NUM 3	FAST lock / release — front antenna / rear antenna.
NUM 7 / NUM 1	Plate lock / release — front reader / rear reader.

NOTE

The six numpad keys are held by the radar whenever the unit is powered, so they will not also trigger a vehicle or menu function. They release as soon as the radar is switched off.

2.2 Commands

Command	Action
/seeker2x_settings	Open the settings menu (also available by holding MENU on the remote).
/seeker2x_move	Reposition and resize the radar display. See section 13.
/prmove2x	Reposition and resize the plate reader overlay.
/seeker2x_windows	Outline and label the five display windows. Useful when lining the unit up against a dashboard.
/toggl doppler2x	Turn the doppler audio on or off.
/togglepr2x	Turn the plate reader on or off.
/seeker2x_radar_debug	Draw the detection beams in the world. Diagnostic only.
/alprlog2x	Print the recent plate lookups to the console.

2.3 Starting the system

#	Step
1	Get into a vehicle the radar is permitted in. By default this is the Emergency class. (<i>server owner</i>)
2	Press F5 . The remote appears. The display head appears with it, dark.
3	Click the PWR pad on the left-hand side of the display head. Power is on the head, not on the remote.
4	The self test runs for about nine seconds and ends on PASS . See section 5.
5	Check the XMIT lamp is lit on each row you intend to use, and that each of those rows has a zone selected.

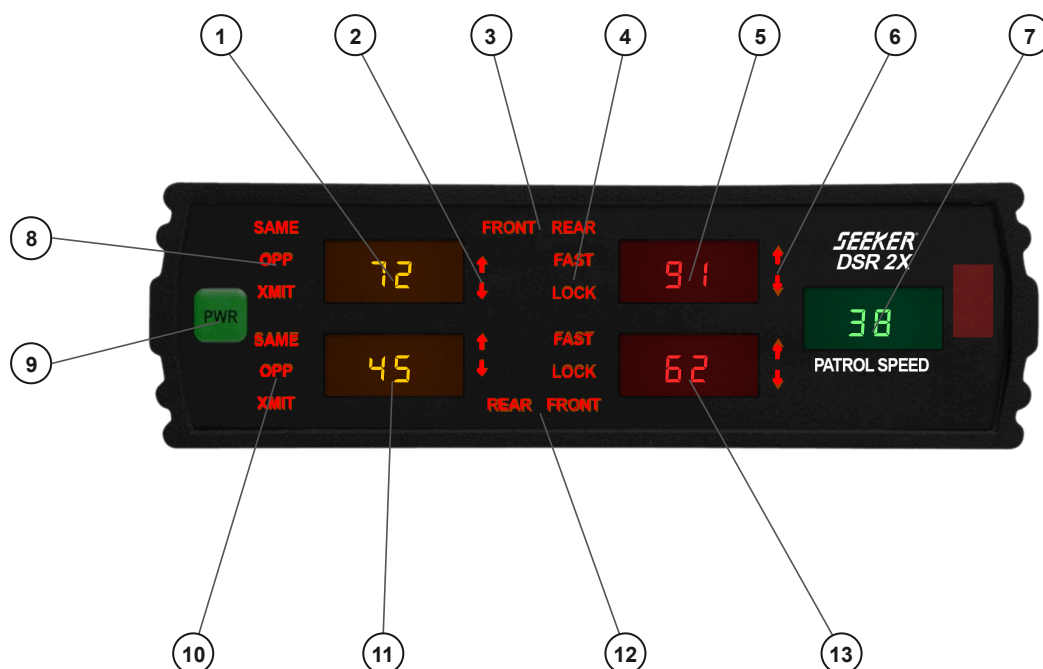
#	Step
6	Set MOV or STA to match how you are working, then set the zones on both rows.

NOTE

The unit always comes up switched off — power is never remembered. Everything else you set *is* remembered, and the first time you switch on after joining the server your saved configuration comes back with it.

3. DISPLAY OPERATION

3.1 Radar display panel



The display unit and its callouts. Every lamp is shown lit; in use only the relevant ones are on.

3.2 Functions of the display unit

#	Item	Description
1	TARGET window, front (amber)	Speed of the strongest target in the front beam.
2	TARGET direction arrows, front	Direction of travel of the vehicle in TARGET.
3	FRONT / REAR lamps, top row	Which antenna owns this row. The top row is always FRONT.
4	FAST / LOCK lamps, top row	Which of the two the red window is currently showing.
5	FAST / LOCK window, front (red)	A faster second speed, or a locked speed.
6	FAST direction arrows, front	Direction of travel of the vehicle in FAST.
7	PATROL SPEED window (green)	Your own speed. Shared by both antennas.
8	SAME / OPP / XMIT lamps, top row	Front antenna's zone selection and transmitter state.
9	PWR pad	Power on and off. Click it while the remote is open.
10	SAME / OPP / XMIT lamps, bottom row	Rear antenna's zone selection and transmitter state.
11	TARGET window, rear (amber)	Speed of the strongest target in the rear beam.
12	FRONT / REAR lamps, bottom row	The bottom row is always REAR. Note the legends are mirrored: on this row REAR is printed on the left.

#	Item	Description
13	FAST / LOCK window, rear (red)	As 5, rear antenna.

3.3 Icon definitions

Lamp	Meaning
XMIT	That antenna's transmitter is on. Unlit means the antenna is switched off or in hold.
SAME	Same-lane zone is selected on that antenna.
OPP	Opposite-lane zone is selected on that antenna.
FRONT / REAR	Which antenna the row belongs to. Fixed labels.
FAST	The red window is showing a faster second target.
LOCK	The red window is showing a locked speed. A lock takes precedence over a FAST reading.

3.4 Arrow direction definitions

Each window has an up arrow and a down arrow beside it. The arrow shows the target's direction of travel *relative to the patrol vehicle*, which is why the same word maps to opposite arrows on the two rows — a vehicle approaching your front bumper and one approaching your rear bumper are travelling opposite ways.

Row	Closing on you	Moving away
Top row (FRONT antenna)	↓	↑
Bottom row (REAR antenna)	↑	↓

NOTE

Neither arrow lights while a target is pacing you. A vehicle sitting alongside at your own speed will show a speed but no arrow, rather than flickering between up and down.

3.5 Patrol window legends

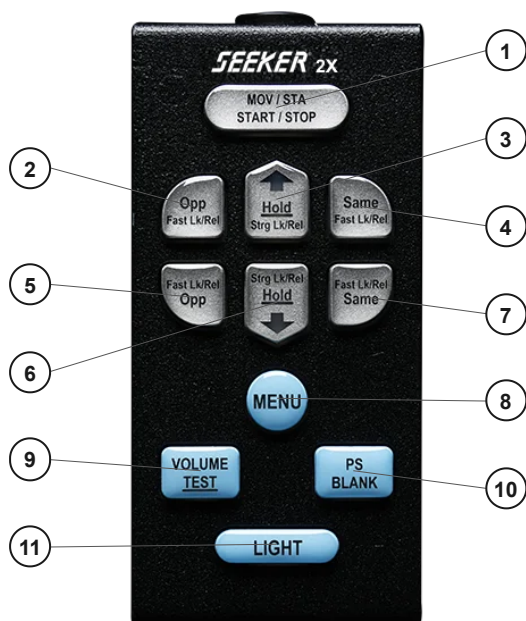
The green window normally shows your own speed, but it is also used to display menu entries and short status messages. You will see the following:

Shows	Meaning
A number	Your patrol speed.
Blank	You are below the PAt cutoff, in stationary mode, or PS BLANK is set to blank.
br1 ... br6	Brightness level, shown briefly after pressing LIGHT.
brA	Automatic brightness.
Menu labels	While in the Operator Menu, VOLUME or Stopwatch Mode, the windows are used for the menu instead of readings.

4. REMOTE CONTROL

The remote carries every operating control except power. Open it with **F5** and close it with **F5** or **ESC**. Buttons only respond while the unit is powered. You can keep driving with the remote open.

4.1 Remote button functions



#	Legend	Function
1	MOV/STA	Moving or stationary mode. START/STOP in Stopwatch Mode.
2	OPP (top)	Front antenna, opposite zone. Once selected, acts as FAST lock/release.
3	STRG (top)	Front antenna speed lock / release. Hold: HLd standby. Moves up in menus.
4	SAME (top)	Front antenna, same zone. Once selected, acts as FAST lock/release.
5	OPP (bottom)	As 2, rear antenna.
6	STRG (bottom)	As 3, rear antenna. Moves down in menus.
7	SAME (bottom)	As 4, rear antenna.
8	MENU	Step through the Operator Menu. Hold: open the settings menu.
9	VOLUME	Step through the volume menu. Hold: run the self test.
10	PS BLANK	Cycle patrol-speed blanking.
11	LIGHT	Light the keypad for six seconds. A second press while it is lit steps the display brightness.

The hand remote. There is no PWR key on the remote — power is on the display head.

4.2 Press and hold

On the real unit an underlined legend means the key has a hold function. On this remote only **STRG** is underlined, and it is the only key with a genuine hardware hold. MENU's and VOLUME's holds are additions for this version.

Key	Press	Hold (½ second)
STRG	Lock / release that antenna's speed	Put that antenna into HLd standby
MENU	Step the Operator Menu	Open the settings menu
VOLUME	Step the volume menu	Run the self test

NOTE

A hold fires the moment the half second passes, not when you let go, and the release is swallowed. You will hear two beeps for a hold and one for a press.

4.3 MOV STA operating modes

MOV/STA changes the whole meaning of the four zone keys. Set the mode *first*, then set the zones.

Mode	Use
MOV	Patrolling. Zone keys select a lane. Patrol speed is subtracted from the reading for you.
STA	Parked and running stationary. Zone keys select a direction of travel. The PATROL window blanks, because there is no patrol speed.

The zone keys are modal

This is the easiest part of the unit to get backwards. The four zone keys do two different jobs depending on whether that zone is already selected:

State of that zone	Lamp	What the key does
Not selected	Dark	Selects the zone.
Already selected	Lit	Acts as FAST lock / release for that antenna.

To **deselect** a zone, press both of that row's zone keys within one second of each other. In stationary mode this is also how you set an antenna bi-directional — pressing both keys together selects both zones rather than toggling one off.

NOTE

In the Operator Menu, the volume menu and Stopwatch Mode, the two **STRG** keys stop being lock keys and become **up** and **down**. Any zone key exits the menu and returns you to normal operation.

HLd standby

Holding **STRG** puts that antenna into HLd. The antenna stops producing readings but keeps its zone selection and stays lit, and any existing lock is kept. Hold STRG again to come out of it. Use HLd rather than switching XMIT off when you only want to pause one antenna briefly.

5. SELF-TEST SEQUENCE

The self test is a fourteen-step diagnostic that exercises every lamp, both antennas' windows, the patrol window and the audio. It runs for about nine seconds and ends on **PASS**.

The keypad is dead for the whole sequence. This is deliberate — a diagnostic you can interrupt halfway is not a diagnostic.

Trigger	When
Power on	Automatically, every time you switch the unit on. (<i>server owner — can be turned off</i>)
Hold VOLUME	Any time the unit is powered.
Timer	Off by default. Real hardware runs one every ten minutes while transmitting. (<i>server owner</i>)

What to watch for

Stage	What should happen
Lamp test	Every lamp on both rows lights, then clears.
Segment test	All five windows run through their digits so you can spot a dead segment.
Audio test	The tone sweeps. If you hear nothing, check the volume settings in section 7.
PASS	Held on screen until the confirmation tone finishes, then the unit returns to normal operation.

NOTE

Existing locks survive the self test and reappear when it finishes.

6. OPERATOR MENU

Press **MENU** to enter. Each press advances to the next step; stepping past the last one exits. Use the two **STRG** keys as up and down to change the value on the current step. Any zone key leaves the menu immediately.

Settings are stored as you change them and survive a restart.

Display	Setting	Values	Notes
FAS	Faster display	On / OFF	Enables the FAST window on both antennas.
OP SEn	Opposite-lane sensitivity	0–4	Detection range for the opposite zone. Applies to both antennas.
SL SEn	Same-lane sensitivity	0–4	Detection range for the same zone. Applies to both antennas.
SqL	Squelch	On / OFF	On, audio is heard only while tracking a target. Off, it free-runs.
PA_t	Patrol speed cutoff	Lo5 / L10 / L20	Below this speed the PATROL window blanks.
StO P	Stopwatch	On / OFF	On, leaving the menu drops you into Stopwatch Mode.
ALE_{rt}	Rear traffic alert	0–99 mph	Closing speed that triggers the alert. 0 disables it.
Ant	Antenna count	1 / 2	Set to 1 to shut down the rear antenna and run as a single-antenna unit.

Sensitivity and range

SEn setting	Approximate range
0	Off — that zone is deaf
1	125 m
2	250 m
3	375 m
4	500 m

Defaults are **OP SEn 4** and **SL SEn 3**. The opposite lane is set further out because oncoming traffic is in the beam for less time. Turn sensitivity down in heavy traffic to stop distant vehicles taking TARGET.

IMPORTANT

While a menu is open the unit **stops acquiring targets** — existing locks are kept, but no new readings are taken. Because of that the menu closes itself: **five seconds with no keypress and the unit returns to radar mode** and starts working again. Each keypress restarts the five seconds, and nothing is lost when it times out, because settings are saved as you change them.

7. VOLUME MENU

Press **VOLUME** to enter, and again to step. Change the value with the two **STRG** keys. Three levels are set independently. Like the Operator Menu, this menu closes itself after five seconds without a keypress.

Display	Controls	Range	Notes
Aud	Doppler tone volume	0–4	Stored separately for moving and stationary mode. You are setting the level for whichever mode you are currently in.
bEE P	Key beeps and alerts	0–3	Also sets the rear traffic alert and the self-test tones.
UOI CE	Voice enunciator	0–3	Set to 0 to silence the spoken confirmations.

The voice enunciator

When you apply a lock the unit speaks the antenna, the mode and the direction — for example "*Front, opposite, closing*". This tells you which antenna produced the reading without having to look away from the road.

NOTE

If the middle word is missing and you hear only two words, the optional voice samples are not installed on the server. The unit works normally; only the middle word is skipped. (*server owner*)

8. STOPWATCH MODE

Stopwatch Mode measures average speed over a marked distance, for use where a doppler reading is not appropriate. Set **StO P** to On in the Operator Menu and leave the menu; the unit drops into Stopwatch Mode.

#	Step
1	Set the measured distance with the STRG up and down keys. The default is 1320 ft — a quarter mile.
2	Press MOV/STA as the vehicle crosses the first mark. The timer starts.
3	Press MOV/STA again as it crosses the second mark. The timer stops.
4	The average speed is displayed. Press any zone key to return to normal operation.

The distance you set is remembered, so a regularly used measured zone only has to be entered once.

NOTE

Holding **STRG** walks the distance value continuously rather than one step per press.

Stopwatch Mode does **not** time out the way the menus do. Waiting between the two marks is the whole point of it, so it stays put until you leave it with a zone key.

9. REAR TRAFFIC ALERT

The rear traffic alert warns you about a vehicle closing on you from behind at speed — the situation you cannot see while you are watching traffic ahead. It is most useful while stopped on a shoulder.

Set the trigger speed on the **ALE rt** step of the Operator Menu. The value is the closing speed in mph. Set it to **0** to disable the alert.

What triggers it

Condition	Requirement
Closing speed	At or above the ALE rt setting.
Still accelerating	The vehicle must still be gaining on you, not merely travelling fast and already slowing.
Range	Within alert range. This works even with SL SEn turned right down, so the alert cannot be accidentally disabled by lowering sensitivity.

What it does

The display flashes and a two-tone warning sounds at the **bEE P** volume. Both stop by themselves once the vehicle is no longer closing. The alert does not interrupt readings and does not clear a lock.

NOTE

The alert works whether or not the rear antenna is transmitting, and whether or not the rear row has a zone selected. It is a safety function, not a measurement.

10. DOPPLER AUDIO

The unit produces a continuously pitched tone that rises and falls with target speed — not a set of stepped tones. With practice you can judge a speed change before the number moves.

What the tone follows

The tone is the audio off the strongest return in the beam, which is the same signal the TARGET reading comes from. It follows **TARGET only**.

Window	Has a tone?
TARGET	Yes. The tone tracks it live.
FAST	No. FAST is a weaker return pulled out of the same signal.
LOCK	No. A lock is a frozen number. Applying a lock does not freeze the tone — the beam is still live, so the audio keeps following whatever is in it.

Both antennas, one speaker

Only one antenna feeds the audio at a time, otherwise two tones would be unusable. By default the unit uses whichever antenna currently has the stronger target. (*server owner — can be fixed to front or rear*)

Overtaking and being overtaken

Situation	Tone
Target faster than you	Tracks the target's true speed.
Target slower than you — you are overtaking it	One steady low tone. Pitching a car you are passing to its road speed would be misleading.
Same-lane traffic behind you	A lower tone based on the speed difference between you, so a car dropping back does not sound like it is speeding up.
Target pacing you	One steady tone. It will not warble across the boundary while you sit alongside each other.

Set the tone volume with **Aud** in the volume menu, and use **Sql** in the Operator Menu to decide whether the audio is silent when nothing is being tracked.

11. BRIGHTNESS AND PS BLANK

11.1 The LIGHT key

LIGHT does two things depending on when you press it:

Press	Result
First press	Lights the remote's keypad for six seconds.
Second press, while the keypad is still lit	Steps the display brightness. The level is shown briefly in the PATROL window as br1 to br6 .

Stepping past level 6 reaches **brA**, automatic. In automatic the display runs bright by day and dims at night. Press again to return to br1.

11.2 PS BLANK

PS BLANK controls what the green PATROL window shows. Press it to cycle through four positions:

Press	Position	Shows
1	Normal	Your current patrol speed.
2	Blank	Nothing. Use when you do not want your own speed on screen.
3	Front lock patrol speed	The patrol speed recorded at the moment the front antenna's lock was taken.
4	Rear lock patrol speed	The same for the rear antenna's lock.

Positions 3 and 4 are the ones that matter in court. Your patrol speed is captured at the instant you lock, so it can be read back afterwards even though you have since slowed, stopped or turned around.

NOTE

Positions 3 and 4 show nothing if that antenna has no lock. The setting is remembered between sessions.

12. PLATE READER AND ALPR

The plate reader is a separate overlay showing the plates of vehicles in front of and behind the patrol vehicle. It is independent of the radar readings and has its own two-slot display. Toggle it with `/togg1epr2x`.

12.1 Manual plate locks

Key	Action
NUM 7	Lock or release the front plate slot.
NUM 1	Lock or release the rear plate slot.

Each slot locks on its own, so you can freeze a plate in front while the rear slot keeps reading, or the other way round. A locked slot stops updating until you release it.

12.2 Continuous ALPR

If the server has ALPR enabled, plates read by the overlay are checked automatically against the CAD. When a record comes back flagged you get an audible hit and the plate is highlighted.

Flag	Meaning
Stolen vehicle	The vehicle is reported stolen.
Impounded vehicle	The vehicle is recorded as impounded.
Expired registration	Registration is missing or not valid.
No insurance	Insurance is missing or invalid.

A plate that has just been checked is not re-checked for five minutes, so sitting behind the same vehicle in traffic will not produce repeated alerts.

NOTE

ALPR is off unless the server owner has enabled it and supplied CAD credentials. Without it the plate reader still shows plates — it simply does not look them up. (*server owner*)

13. MOVING AND SCALING THE INTERFACE

All three on-screen elements can be positioned and sized independently, and each remembers its position between sessions.

Element	Command	Notes
Radar display head	<code>/seeker2x_move</code>	The whole unit scales together — face, lamps and digits.
Plate reader overlay	<code>/prmove2x</code>	
Remote control	Its own adjust mode	Reached from the settings menu.

#	Step
1	Run the command. The element becomes draggable and a small guide appears.
2	Drag to move it.
3	Scroll the mouse wheel over it to resize.
4	Press <code>ESC</code> or <code>ENTER</code> to save and exit.

Use `/seeker2x_windows` to outline and label the five windows. This makes it easy to line the unit up against a dashboard or a specific part of the screen.

If you lose the display off-screen, reset it to its default position from the settings menu (`/seeker2x_settings`).

14. TROUBLESHOOTING

Symptom	Check
No readings on either row	Is the unit powered? Is XMIT lit? Is at least one zone selected on that row? Is SEn for that zone set to 0?
No readings on the bottom row only	Ant in the Operator Menu is set to 1, which shuts down the rear antenna. Set it to 2.
Row is lit but produces nothing	That antenna is in HLd . Hold its STRG key to come out of standby.
Readings stopped while I was setting something	A menu is open. The unit does not acquire targets while a menu is on screen. Press a zone key to exit, or just wait — it closes itself after five idle seconds.
The menu closed on its own	Five seconds passed with no keypress. This is deliberate, so a forgotten menu cannot leave the radar deaf. Your settings were kept.
Zone key locks FAST instead of selecting the zone	The zone is already selected. Press both of that row's zone keys within a second to deselect.
Arrows point the wrong way	They are relative to the patrol vehicle and are mirrored between the two rows. See section 3.4.
No audio	Check Aud in the volume menu, check SqL , and check doppler is not switched off with /toggledoppler2x.
Tone does not stop when I lock	Correct behaviour. A lock freezes the number, not the beam.
Voice says only two words	The optional voice samples are not installed on the server. (<i>server owner</i>)
PATROL window is blank	You are below the PAt cutoff, in stationary mode, or PS BLANK is set to blank.
Remote will not respond	The unit is not powered. Click the PWR pad on the display head, not the remote.
Keypad is dead for a few seconds	The self test is running. Wait for PASS.
Radar will not appear at all	You are not in a permitted vehicle. By default only Emergency class vehicles are allowed. (<i>server owner</i>)
Display is off-screen	Reset it to defaults from /seeker2x_settings.